Cetuximab in combination with 5-fluorouracil, leucovorin and irinotecan as a neoadjuvant chemotherapy in patients with initially unresectable colorectal liver metastases.

PMID: 18063876 | Indexed in PubMed

The efficacy and safety of a combination of cetuximab, irinotecan, and 5-fluorouracil/leucovorin (FOLFIRI) in downsizing unresectable colorectal liver metastases was investigated. Patients with unresectable colorectal liver metastases with or without resectable extrahepatic metastasis were enrolled. 23 patients initially received 400 mg/m2 of cetuximab, followed by a weekly infusion of 250 mg/m2 and a biweekly dose of irinotecan (180 mg/m2), with 5-fluorouracil both by bolus (400 mg/m2) and by a 46-h infusion (total of 2,400 mg/m2) with leucovorin (400 mg/m2). The overall response rate was 39.1% (n = 9; 95% confidence interval (CI): 17.6-60.7%). The most common grade 3-4 toxicities were skin reactions (30.4%) and diarrhea (26.1%). The rate of conversion to resectable liver metastases was 30.4% (n = 7; 95% CI: 10.1-50.8%). The factors found to be significantly associated with R0 resection were lower serum carcinoembryonic antigen levels after chemotherapy (p = 0.039), being chemonaive (p = 0.002), and showing a higher incidence of grade 3-4 skin toxicity (p = 0.011). Cetuximab with FOLFIRI may be an effective and safe treatment option for downsizing unresectable colorectal liver metastases.

Neoadjuvant and conversion treatment of patients with colorectal liver metastasis: the potential role of bevacizumab and other antiangiogenic agents.

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More than 50 % of patients with colorectal cancer develop liver metastases. Surgical resection is the only available treatment that improves survival in patients with colorectal liver metastases (CRLM). New antiangiogenic targeted therapies, such as bevacizumab, aflibercept, and regorafenib, in combination with neoadjuvant and conversion chemotherapy may lead to improved response rates in this population of patients and increase the proportion of patients eligible for surgical resection. The present review discusses the available data for antiangiogenic targeted agents in this setting. One of these therapies, bevacizumab, which targets the vascular endothelial growth factor (VEGF) has demonstrated good results in this setting. In patients with initially unresectable CRLM, the combination of 5-fluorouracil, leucovorin, and oxaliplatin (FOLFOX) plus bevacizumab has led to high response and resection rates. This combination is also effective for patients with unresectable CRLM. Moreover, the addition of bevacizumab to chemotherapy in the neoadjuvant setting of liver metastasis has a higher impact on pathological response rate. This drug also has a manageable safety profile, and according to recent data, bevacizumab may protect against the sinusoidal dilation provoked in the liver by certain cytotoxic agents. In phase II trials, antiangiogenic therapy has demonstrated benefits in the presurgical treatment of CRLM and may represent a new treatment pathway for these patients.

[Neoadjuvant chemotherapy or primary surgery for colorectal liver metastases. Pro adjuvant chemotherapy].

PMID: 24292192 | Indexed in PubMed

Medicinal treatment of colorectal cancer liver metastases has undergone significant developments in the past two decades. Cytotoxic treatment regimens have demonstrated greater efficacy and contributed to a significant improvement in survival. Perioperative therapy with FOLFOX has demonstrated progression-free survival benefits in resectable colorectal liver metastases in a prospective randomized controlled trial. The safety of perioperative chemotherapy was found to be acceptable and only a few patients showed initial progression. In general, preoperative chemotherapy should not extend beyond 3-4 months to avoid chemotherapy-associated liver damage. For patients with borderline resectable metastases, a response to chemotherapy can establish secondary resectability. Combinations of cytotoxic drugs are adequate in this situation. The addition of bevacizumab or anti-epidermal growth factor receptor (EGFR) antibodies for tumors without RAS mutations are reasonable choices for a better response and improved survival outcome. It is crucial that all treatment decisions for systemic colorectal liver metastases include correct definitions of treatment goals and endpoints and are derived based on appropriate multidisciplinary considerations.

[Treatment of colorectal cancer liver metastases].

PMID: 16565703 | Indexed in PubMed

Liver is the main target for colorectal cancer (CRC) metastases. About 50% of all patients affected by CRC develop liver metastases. Surgery remains the only potentially curative strategy and indications to surgery and resectability criteria are now less restrictive than before so that a more aggressive attitude in the treatment of metastatic lesions is the rule. However surgery is not possible in the majority of patients. For non resectable patients two options are available: local treatment strategies (Radio-frequency ablation and Cryosurgery: alone or in combination with surgery) and chemotherapy. High rates of objective response achieved with Fluoropyrimidines, Oxaliplatin (OHP) and Irinotecan (CPT-11) based chemo-therapy, enable initially unresectable patients to undergo surgery, with a 5-year survival rate comparable to that observed for primary resectable patients. Therefore chemotherapy has not only a palliative aim, but becomes a fundamental moment of a combined medical and surgical treatment with curative purpose. After surgery two-thirds of patients will relapse in first two years, so that adjuvant therapy has been investigated to reduce recurrence rates, mainly testing hepatic arterial infusion (HAI) schedules. So far no randomized trials have been published on the role of systemic intravenous adjuvant chemo-therapy. Finally we report the results of our monoinstitutional experience, suggesting a possible role of systemic adjuvant chemotherapy in reducing recurrence rates after liver metastasectomy. Probably in the next years new targeted drugs and locoregional therapies will contribute to further improve prognosis of such patients, in a neoadjuvant, adjuvant and palliative setting.

[Application of molecular targeted agents in comprehensive treatment of gastrointestinal cancer].

PMID: 25421761 | Indexed in PubMed

Targeted agents increase response rates and improved overall survival in treatment of metastatic gastrointestinal cancer. Therefore, physicians pay more attention to the role of targeted agents in treatment of local advanced gastrointestinal cancer. The clinical trials are ongoing to evaluate the efficacy of Trastuzumab in neoadjuvant treatment of local advanced gastric cancer with HER-2 gene over expression. Many studies reported Cetuximab plus chemotherapy as a conversion treatment improve R0 resection rates and prolonged overall survival of the patients with potentially resectable colorectal cancer liver metastasis with wild type KRAS gene status. A phase III(clinical trial is assessing the conversion efficacy of Bevacizumab in unresectable disease with KRAS gene mutation. Current evidence showed that neoadjuvant therapy of targeted agents did not prolong survival of patients with resectable liver metastasis. However, this is controversial. In neoadjuvant therapy of local advanced rectal cancer, Cetuximab did not improve the rates of pathological complete response in most of the phase II(trials. Furthermore, there are no phase III(trials to assess the role of Bevacizumab. Compared to chemotherapy alone for metastatic cancer, it is more important to evaluate the interaction and synergistic action of targeted agents, cytotoxic drugs, surgery and radiation, to make a scientific multidisciplinary model in comprehensive treatment of local advanced cancer.

[Therapeutic effect of mFOLFOX6 for synchronous unresectable liver metastases from colorectal cancer].

PMID: 22202333 | Indexed in PubMed

The current chemotherapy for metastatic colon cancer has improved an overall survival. In this study, we retrospectively analyzed the efficacy of mFOLFOX6 in colorectal cancer patients with synchronous unresectable liver metastases and compared the prognosis between before and after the administration of mFOLFOX6. The subject was 28 patients of colorectal cancer with synchronous unresectable liver metastasis who received mFOLFOX6 as a first-line treatment from 2005 to 2010. The median frequency of mFOLFOX6 was 10 times(range, 2-24 times), relative dose intensity of oxaliplatin was 75.0% (range, 42 .9-100), response rate was 32%, and median progression-free survival was 9 . 9 months. Surgical resection of colorectal liver metastases was performed to 4 patients (14.3%) as a conversion therapy. The overall survival of the patients with mFOLFOX6 was significantly better than that of 31 patients who received the chemotherapy via hepatic artery or the chemotherapy before the administration of oxaliplatin (31.8 months vs. 15 .1 months, $p<0.01$). Our results suggested that mFOLFOX6 treatment for unresectable liver metastases of colorectal cancer was made not only the conversion therapy possible, but it has improved the prognosis when compared with previous treatment without oxaliplatin.

[Consideration of therapy for colorectal cancer with synchronous unresectable liver metastasis].

PMID: 23980038 | Indexed in PubMed

A variety of managements, including systemic and local chemotherapy, radiofrequency ablation and others, are used after multidisciplinary team discussion to improve the survival of patients with unresectable liver metastasis, and to enlarge the cohort of patients who can be managed with curative intent. Patients should be divided into different clinical groups according to characteristics of the patient and tumor, and then receive different treatments. For the patients who may be converted to be resectable after chemotherapy, we should choose efficient convertible chemotherapy with short courses to get the best response rate. For KRAS wild-type patients, cetuximab combined with FOLFOX/FOLFIRI, in which 5-fluorouracil is continuously infused, is recommended. In addition, resection of the primary tumor is recommended at the right time for asymptomatic patients with unresectable liver metastases. There is no consensus on the preferred treatment modality for systemic and local therapies.

Modified FOLFOXIRI With or Without Cetuximab as Conversion Therapy in Patients with RAS/BRAF Wild-Type Unresectable Liver Metastases Colorectal Cancer: The FOCULM Multicenter Phase II Trial.

PMID: 33400355 | Indexed in PubMed

This trial evaluated the addition of cetuximab to a modified FOLFOXIRI (mFOLFOXIRI: 5-fluorouracil/folinic acid, oxaliplatin, irinotecan) as conversion therapy in a two-group, nonrandomized, multicenter, phase II trial in patients with initially technically unresectable colorectal liver-limited metastases (CLM) and BRAF/RAS wild-type. Patients were enrolled to receive cetuximab (500 mg/m2) plus mFOLFOXIRI (oxaliplatin 85 mg/m2 , irinotecan 165 mg/m2 , folinic acid 400 mg/m2 , 5-fluorouracil 2,800 mg/m2 46-hour infusion, every 2 weeks) (the cetuximab group) or the same regimen of mFOLFOXIRI alone (the control group), in a 2:1 ratio allocation. The primary endpoint was the rate of no evidence of disease (NED) achieved. Secondary endpoints included resection rate, objective response rate (ORR), survival, and safety. Between February 2014 and July 2019, 117 patients were registered for screening at six centers in China, and 101 of these were enrolled (67 cetuximab group, 34 control group). The rate of NED achieved was 70.1% in the cetuximab group and 41.2% in the control group (difference 29.0%; 95% confidence interval [CI], 9.1%-48.8%; p = .005). Patients in the cetuximab group had improved ORR (95.5% vs. 76.5%; difference 19.1%; 95% CI, 17.4%-36.4%; p = .010) compared with those in control group. Progression-free survival and overall survival showed the trend to favor the cetuximab group. The incidence of grade 3 and 4 adverse events was similar in the two groups. Addition of cetuximab to mFOLFOXIRI improved the rate of NED achieved. This combination could be an option of conversion regimen for molecularly selected patients with initially technically unresectable CLM. This trial evaluated the addition of cetuximab to a modified FOLFOXIRI as conversion therapy in a phase II trial in patients with initially technically unresectable colorectal liver-limited metastases and BRAF/RAS wild-type. The rate of no evidence of disease achieved was 70.1% in the cetuximab plus modified FOLFOXIRI group and 41.2% in the modified FOLFOXIRI group. Objective response rates, overall survival, and progression-free survival were improved in the cetuximab group when compared with the modified FOLFOXIRI group. Addition of cetuximab to modified FOLFOXIRI increased the rate of no evidence of disease achieved, and this combination could be an option of conversion regimen for molecularly selected patients with initially technically unresectable colorectal liver-limited metastasis.

Chemotherapy (doublet or triplet) plus targeted therapy by RAS status as conversion therapy in colorectal cancer patients with initially unresectable liver-only metastases. The UNICANCER PRODIGE-14 randomised clinical trial.

PMID: 34992255 | Indexed in PubMed

Colorectal cancer (CRC) patients have a better prognosis if metastases are resectable. Initially, unresectable liver-only metastases can be converted to resectable with chemotherapy plus a targeted therapy. We assessed which of chemotherapy doublet (2-CTx) or triplet (3-CTx), combined with targeted therapy by RAS status, would be better in this setting. PRODIGE 14 was an open-label, multicenter, randomised Phase 2 trial. CRC patients with initially defined unresectable liver-only metastases received either, 2-CTx (FOLFOX or FOLFIRI) or 3-CTx (FOLFIRINOX), plus bevacizumab/cetuximab by RAS status. The primary endpoint was to increase the R0/R1 liver-resection rate from 50 to 70% with the 3-CTx. Patients (n = 256) were mainly men with an ECOG PS of 0, and a median age of 60 years. In total, 109 patients (42.6%) had RAS-mutated tumours. After a median follow-up of 45.6 months, the R0/R1 liver-resection rate was 56.9% (95% CI: 48-66) with the 3-CTx versus 48.4% (95% CI: 39-57) with the 2-CTx (P = 0.17). Median overall survival was 43.4 months with 3-CTx versus 40 months with 2-CTx. We failed to increase from 50 to 70% the R0/R1 liver-resection rate with the use of 3-CTx combined with bevacizumab or cetuximab by RAS status in CRC patients with initially unresectable liver metastases.

[Late-onset schizophrenia and schizophrenia-like psychosis with very late onset].

PMID: 39113441 | Indexed in PubMed

This review focuses on late-onset schizophrenia and schizophrenia-like psychosis with very late onset (VLOSLP) with focus on their psychopathologic, neuropsychologic, and neurobiologic aspects. A literature review on late-onset schizophrenia and VLOSLP was conducted based on publications from PubMed, Scopus, and Google Scholar databases up to December 2023. It may be noted that research into schizophrenia has largely focused on early-onset patients, and research into the mental health of older people has focused primarily on dementia and depression, with relatively little information on late-onset schizophrenia and VLOSLP. The nosology of late-onset functional psychoses is still poorly understood. There is currently no consensus on the diagnostic framework for psychosis labeled by the term VLOSLP. These deficiencies need to be addressed in order to understand the background of VLOSLP, the course and prognosis of the illness, and to develop successful management and treatment strategies for these patients, as older adults are more susceptible to the adverse effects of psychotropic medications. Therapy should be holistic, including not only medication but also psychotherapy, and the key role of caregivers of elderly schizophrenia patients should be taken into account. There should be judicious use of pharmacotherapy with an assessment of its risks and benefits.